

Indian Institute of Technology Dharwad
CS201L: Artificial Intelligence Laboratory
Lab 3: Data Cleaning - Handling Missing Values
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Problem Statement 2

A real-world medical dataset containing hospital appointment records is provided in a CSV file (`noshowappointments.csv`). The dataset includes various attributes that provide insights into patient appointment behavior. These features are:

Table 1: Description of Attributes in the Patient Appointment Dataset

Feature	Description
PatientId	Unique identifier assigned to each patient.
AppointmentID	Unique identifier assigned to each appointment record.
Gender	Gender of the patient.
ScheduledDay	Date on which the appointment was scheduled.
AppointmentDay	Actual date of the appointment.
Age	Age of the patient in years.
Neighbourhood	Residential area where the patient lives.
Scholarship	Indicates whether the patient is enrolled in a social welfare program.
Hypertension	Indicates whether the patient has hypertension.
Diabetes	Indicates whether the patient has diabetes.
Alcoholism	Indicates whether the patient has a history of alcoholism.
SMS_received	Indicates whether the patient received a reminder SMS.
No-show	Indicates whether the patient failed to attend the appointment.

Write a Python program to perform the following tasks:

1. Display the number of missing values in each attribute. Also, find the total number of missing values in the file.
2. For the file `noshowappointments.csv`:
 - (a) Count and display the number of tuples having one, two, three, four, or upto six missing values. Plot a graph for the **number of missing values** vs. the **number of tuples**.
 - (b) Count and display the number of tuples having equal to or more than 50% of attributes with missing values.

- (c) Delete (drop) the tuples equal to or more than 50% of attributes with missing values.
- (d) The target (class) attribute is **No-show**. Drop the tuple missing in the target (class) attribute.
- (e) Then save updated dataset after doing steps (c) and (d) as `noshowappointments_updated.csv`

Instruction: Create a new Jupyter Notebook and load the updated CSV file obtained after deleting the required tuples in the previous step. All further analysis must be performed using this updated dataset.

3. Count and display the number of missing values in each attribute. Also, find the total number of missing values in the file (after the deletion of tuples).
4. For the file `noshowappointments_updated.csv` - Experiments on Filling Missing Values
 - (a) Compute the mean, median, mode, and standard deviation for each numerical attribute for the file `noshowappointments_updated.csv`.
 - (b) Replace the missing values for numerical attributes with the median of their respective attributes. (Use `df.fillna()` with suitable arguments.) Count and display the number of missing values in each attribute after replacing missing values.
 - (c) Compute the mean, median, mode, and standard deviation for each numerical attribute after replacing the missing values with median and observe the changes.
 - (d) Replace the missing values in each numerical attribute by propagating the previous non-missing value (forward padding). (Use `df.fillna()` with suitable arguments). Count and display the number of missing values in each attribute after replacing missing values.
 - (e) Replace the missing values in each numerical attribute by propagating the next non-missing value (backward padding). (Use `df.fillna()` with suitable arguments). Count and display the number of missing values in each attribute after replacing missing values.
 - (f) Replace the missing values in each numerical attribute using the linear interpolation technique. (Use `df.interpolate()` with suitable arguments). Count and display the number of missing values in each attribute.